

Generation Interconnection Request System Impact Study Report

For Queue Project Q14-0388

Old Hickory 138 kV
June 15, 2016

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q14-0388. The Interconnection Customer proposes a 82.2 MW solar photovoltaic facility interconnecting at the Old Hickory 138 kV substation in the Virginia area. The request was received on October 01, 2014. The study evaluated system intact and single contingency conditions using the regional planning case, including 193 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 10 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$1,089,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Fritch-Big Hanaford 138kV
Contingency: Mitchell Dam-Skaggs Island Jct 2 345kV
Loading: 114.2% (Rating: 281 MVA)
DFAX: 10.3%
Pre-project loading: 111.2%
2. Facility: Dist 1000-Birchwood 138kV
Contingency: Bear Island-Dunn Siding 345kV Line
Loading: 110.0% (Rating: 304 MVA)
DFAX: 5.7%
Pre-project loading: 108.5%
3. Facility: Carver-Shockoe 138kV
Contingency: Plaza Substation-Big Hanaford 138kV line
Loading: 109.1% (Rating: 230 MVA)
DFAX: 5.8%
Pre-project loading: 107.1%
4. Facility: Elko-Syl Laskin 138kV
Contingency: Mtchell Dam-Haggin 345kV
Loading: 109.0% (Rating: 200 MVA)
DFAX: 6.2%
Pre-project loading: 106.5%
5. Facility: Trenton Diesel-Dist 1000 138kV
Contingency: Bear Island Substation-Dunn Siding 345kV Line
Loading: 106.2% (Rating: 325 MVA)
DFAX: 5.7%
Pre-project loading: 104.8%

6. Facility: Drum-Montplier 138kV
Contingency: Dixleta-Taber 345kV L
Loading: 105.2% (Rating: 432 MVA)
DFAX: 5.3%
Pre-project loading: 104.2%
7. Facility: Westport Substation-Wheelabrator Baltimore Refuse 138kV
Contingency: Taber-Dixleta 345kV
Loading: 102.2% (Rating: 589 MVA)
DFAX: 5.7%
Pre-project loading: 101.4%
8. Facility: Dixleta-North Pole 138kV Ckt 1
Contingency: Taber Substation-Glenannan 345kV line
Loading: 101.7% (Rating: 465 MVA)
DFAX: 5.3%
Pre-project loading: 100.7%
9. Facility: Bixby Hill-Tombstone 500kV Line
Contingency: Sharyland-Tombstone 345kV L
Loading: 101.0% (Rating: 3,389 MVA)
DFAX: 8.8%
Pre-project loading: 100.8%
10. Facility: Carver-Big Hanaford 138kV
Contingency: Plaza-Big Hanaford 138kV
Loading: 100.2% (Rating: 187 MVA)
DFAX: 5.8%
Pre-project loading: 97.6%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Big Hanaford-Fritch 138kV	Rebuild line with higher-capacity conductor	\$4,345,000	2% (\$90,000)
Dist 1000-Birchwood 138kV	Reconductor line	\$8,951,000	2% (\$169,000)
Carver-Shockoe 138kV	Reconductor line	\$505,000	4% (\$19,000)
Elko-Syl Laskin 138kV	Reconductor line	\$1,658,000	2% (\$30,000)
Trenton Diesel-Dist 1000 138kV Line	Reconductor line	\$14,532,000	2% (\$275,000)
Drum-Montpelier Substation 138kV	Reconductor line	\$2,761,000	1% (\$31,000)
Westport Substation-Wheelabrator Baltimore Refuse 138kV	Reconductor line	\$211,000	1% (\$2,000)
Dixleta-N Pole 138kV	Reconductor line	\$468,000	1% (\$4,000)
Bixby Hill-Tombstone 500kV Line	Reconductor line	\$59,211,000	1% (\$446,000)
Carver-Big Hanaford 138kV	Reconductor line	\$602,000	4% (\$23,000)
	Total allocated to project		\$1,089,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.